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# A Spectroscopic Redshift Catalog and Dynamical Substructure Analysis of Strong-Lensing-Selected Galaxy Clusters from the SDSS Giant Arcs Survey

## ABSTRACT

We present a spectroscopic redshift catalog and a dynamical substructure analysis of 54 strong-lensing-selected galaxy clusters from the SDSS Giant Arcs Survey (SGAS). We obtained Magellan/IMACS multi-object spectroscopy in six SGAS cluster fields and measured 980 redshifts with Marz quality QOP $\geq 2$ (912 with QOP $= 4$). Combined with literature and NED redshifts, these form a catalog of 7,543 unique objects in 74 cluster fields. Using a velocity pre-filter and iterative $3\sigma$ biweight clipping, we measure redshifts, velocity dispersions, and dynamical masses for the 54 clusters with at least 15 members. For the 35 clusters with at least 30 members, we test for substructure with a Monte-Carlo-calibrated Dressler–Shectman (DS) test, verified with Gaussian mixture models, and detect it ($p_{\rm DS} \leq 0.05$) in 13. Because this membership is based on velocity alone, it includes galaxies 4–6 Mpc from the cluster centers. Restricting members to within $R_{200}$ lowers the substructure fraction from $0.37 \pm 0.08$ to $0.15 \pm 0.07$ ($0.11 \pm 0.06$ within 1.5 Mpc), and simulated mergers show that this drop is not caused by the smaller samples. Kinematic and X-ray dynamical-state classifications agree for only 5 of 16 overlap clusters (31%), with or without the $R_{200}$ restriction. Inside $R_{200}$, none of the 10 X-ray-disturbed clusters shows kinematic substructure, which argues against large line-of-sight velocity offsets in their mergers. SDSS J1152+0930 is the only cluster whose galaxy velocities show substructure at every aperture while its X-ray image looks relaxed, making it our best candidate for a merger along the line of sight, which can leave the X-ray image looking undisturbed.

## 1. INTRODUCTION

Galaxy clusters are the most massive gravitationally bound structures in the universe and constitute a powerful laboratory for both astrophysics and cosmology. Their abundance as a function of mass and redshift constrains cosmological parameters (e.g. Allen et al. 2011), their density profiles trace the dark matter halo (Navarro et al. 1996), and their dynamical state encodes the recent history of structure assembly (Kravtsov & Borgani 2012).

A cluster's dynamical state, whether it is fully virialized ("relaxed") or still merging ("disturbed"), is inferred from observational signatures across multiple wavelengths. In the optical, kinematic substructure tests such as the Dressler–Shectman (DS) statistic (Dressler & Shectman 1988) probe deviations of local galaxy kinematics from the global cluster distribution. In X-rays, the morphology of the intracluster medium (ICM), through indicators such as concentration, asymmetry, the centroid shift, and the X-ray–BCG offset (Rasia et al. 2013; Cialone et al. 2018), traces gas disturbances on merger timescales. Because galaxies and ICM respond to mergers on different timescales and along different projections, multi-wavelength dynamical-state classifications are more informative than any single tracer (e.g. Foëx et al. 2017).

Strong-lensing-selected cluster samples are a particularly interesting population in this context. Selection on the presence of giant arcs preferentially identifies cluster cores with high projected central mass densities, and the question of whether this selection biases the population toward unusual dynamical states is not fully settled (Meneghetti et al. 2014; Gassis et al. 2026). The SDSS Giant Arcs Survey (SGAS; Hennawi et al. 2008; Gladders et al. 2013; Bayliss et al. 2014; Sharon et al. 2020) has assembled a homogeneous sample of several dozen strong-lensing clusters, with HST follow-up, Chandra imaging for the brighter subsample, and published strong-lensing mass models for the cores. However, spectroscopic completeness in the field surrounding each cluster has historically been heterogeneous, limiting the statistical power of any kinematic dynamical-state analysis on this sample.

In this paper we (i) present new IMACS/Magellan multi-object spectroscopy in the fields of 6 SGAS clusters, re-measured from scratch with the Marz redshifting tool to yield 980 redshifts with QOP $\geq 2$; (ii) merge these measurements with literature redshifts into a homogenized master catalog of 9,176 pre-deduplication en-

tries (7,543 unique objects after cross-matching) across 74 cluster fields; (iii) develop an end-to-end pipeline for cluster membership selection, biweight dynamics with jackknife 95% confidence intervals, the DS substructure test with seeded Monte-Carlo calibration, and GMM-based substructure verification; (iv) apply this pipeline to the 35 DS-eligible clusters ($N \geq 30$) and report virial-mass and substructure results for each of the 54 dynamics-eligible ($N \geq 15$) clusters; and (v) compare our kinematic dynamical-state indicators to the X-ray morphological classifications of the Chandra Strong Lens Sample (Gassis et al. 2026), which overlaps our sample; and (vi) test how the substructure fraction and the X-ray comparison depend on the spatial aperture used to define membership, calibrating the detection power of the DS test with mock mergers. Preliminary results based on a smaller subsample were presented in Adhikari et al. (2024).

Throughout this paper we assume a flat ΛCDM cosmology with $H_0 = 67.66\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.3097$ ($\Omega_\Lambda = 1 - \Omega_m$), the Planck Collaboration et al. (2020) parameters as implemented in `astropy`.

## 2. SAMPLE AND OBSERVATIONS

### 2.1. *The SGAS strong-lensing cluster sample*

Our target list is drawn from the SGAS strong-lensing cluster sample of Sharon et al. (2020), which provides published strong-lensing mass models for 37 clusters spanning $0.16 < z < 0.6$. The sample was originally identified by visual inspection of SDSS imaging for arc-like features, with follow-up imaging confirming the lensing geometry (Hennawi et al. 2008; Gladders et al. 2013; Bayliss et al. 2014). Einstein radii (for a source at $z = 2$) range from $1\farcs3$ to $23\farcs1$, with a median of $10\farcs8$ (Sharon et al. 2020); the corresponding core projected masses are typically $10^{14} M_\odot$, indicating that the sample is dominated by massive systems.

For the present work we focus on the subset of SGAS clusters that either have new IMACS multi-object spectroscopy from our program or have substantial archival spectroscopic redshift coverage from the literature, yielding 74 fields with at least one spectroscopic redshift, 54 clusters meeting the $N \geq 15$ dynamics threshold of Beers et al. (1990), and 35 clusters meeting the $N \geq 30$ Dressler–Shectman threshold of Pinkney et al. (1996).

### 2.2. *IMACS multi-object spectroscopy*

We obtained multi-object spectroscopy of galaxies in the fields of 6 SGAS clusters using the Inamori-Magellan Areal Camera and Spectrograph (IMACS; Dressler et al. 2011) on the Magellan-I (Baade) telescope at Las Campanas Observatory, over several observing runs in 2013–2014. Observations used the IMACS $f/2$ short camera ($0\farcs200$ pixel$^{-1}$ plate scale) with the 200 line mm$^{-1}$ grism (`Gri-200-15.0`) in multi-slit mode, covering a wavelength range of ≈ 5000–9500 Å dispersed at ≈ 1 Å pixel$^{-1}$, with slit widths of ≈ $1''$ and a CCD binning of $1 \times 1$. A total of 9 multi-slit masks were observed across the six fields, with typical per-mask on-sky integrations of ∼40 min (∼2400 s). A summary of the observations is given in Table 1.

**Table 1.** IMACS/Magellan multi-object spectroscopic observations

| Cluster | $N_{\mathrm{masks}}$ | $N_{\mathrm{spec}}$ | Exp. (s) | UT date |
|---|---|---|---|---|
| SDSS J0146−0929 | 1 | 147 | 2400 | 2013 Nov 07 |
| SDSS J1152+0930 | 2 | 338 | 2400 | 2014 Apr 25–26 |
| SDSS J1336−0331 | 2 | 278 | 2400 | 2014 Apr 25–26 |
| SDSS J1439+1208 | 1 | 144 | 1600 | 2013 Apr 09 |
| SDSS J1621+0607 | 1 | 251 | 2400 | 2013 Aug 04 |
| SDSS J2243−0935 | 2 | 281 | 2400 | 2013 Jul 12; Nov 07 |
| Total | 9 | 1439 | | |

NOTE—$N_{\mathrm{spec}}$ is the number of one-dimensional spectra extracted from each field's masks. Exposure is the typical per-mask on-sky integration; UT dates list the night(s) each field's masks were observed with the IMACS $f/2$ short camera and the 200 line mm$^{-1}$ grism.

### 2.3. *Data reduction*

Two-dimensional spectra were reduced using the `COSMOS` pipeline[1] (Kelson 2003), which performs bias subtraction, flat-fielding using internal quartz-lamp exposures, wavelength calibration against HeNeAr arc lamps, sky subtraction using off-target rows within each slitlet, and relative flux calibration using standard-star observations. Cosmic-ray rejection was performed via L.A. Cosmic (van Dokkum 2001). We then optimally extracted one-dimensional spectra from each cleaned slitlet following the method of Horne (1986), with the extraction aperture matched to the spatial extent of each object's continuum trace. Example two-dimensional and extracted one-dimensional spectra are shown in Figure 1.

## 3. REDSHIFT MEASUREMENTS

Redshifts were measured in two passes. The first pass used cross-correlation under IRAF; we then re-measured every spectrum from scratch with the MARZ redshift-

[1] https://code.obs.carnegiescience.edu/cosmos

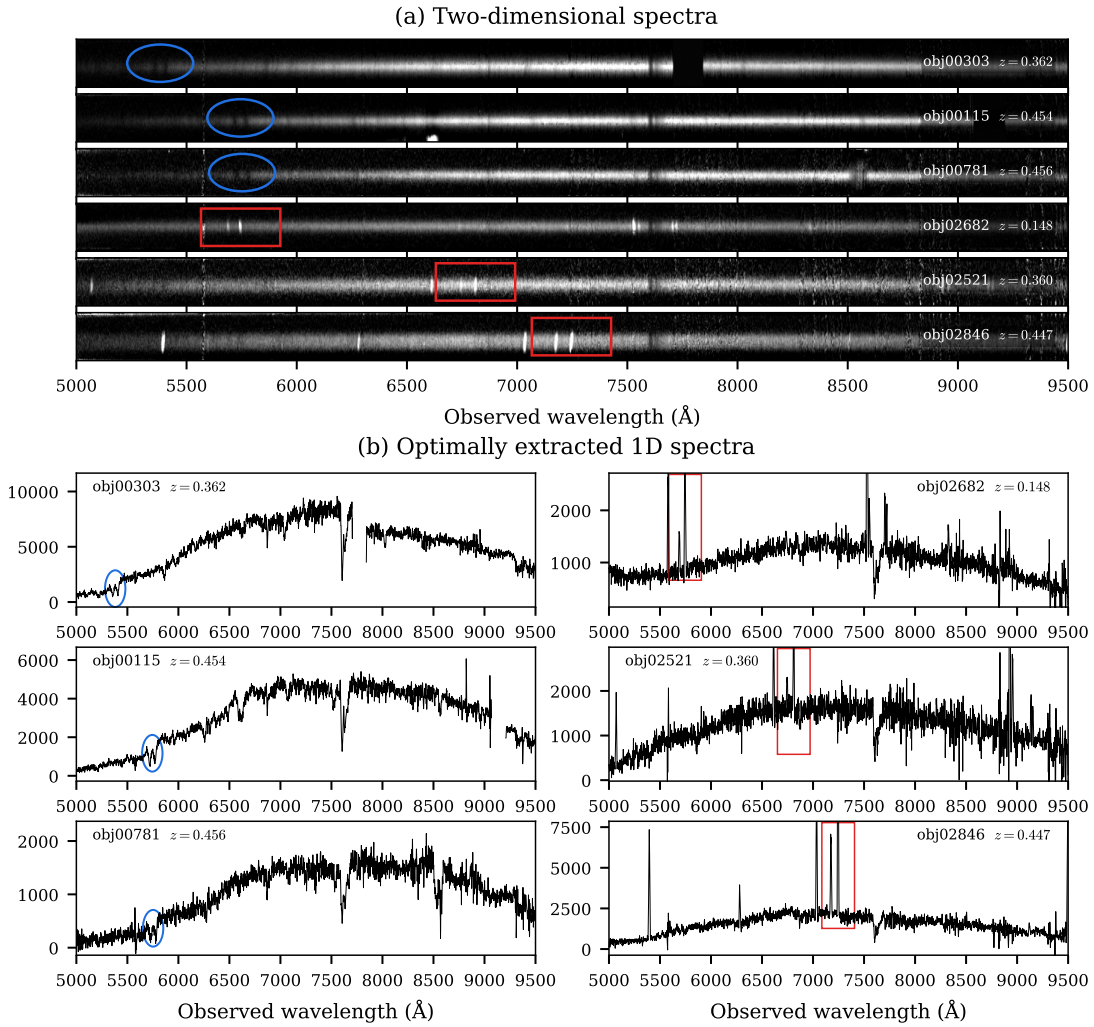



**Figure 1.** Example IMACS/Magellan spectra of six member galaxies in the field of SDSS J2243−0935 (mask 2). *(a)* The reduced two-dimensional spectra; *(b)* the corresponding optimally extracted one-dimensional spectra, with absorption-line members in the left column and emission-line members in the right column. Blue ellipses mark the redshifted Ca II H&K doublet used for absorption-line cross-correlation, and red rectangles mark the strongest redshifted nebular emission ([O III] $\lambda$5007; [O II] $\lambda$3727 also falls in band for the higher-redshift emission galaxies). All feature positions are computed from each object's MARZ redshift (labelled in each panel), so they coincide with the observed lines.

ing tool. The adopted redshifts draw on both passes: for the large majority of objects the IRAF and MARZ redshifts agree closely and are treated as mutually validated, while MARZ additionally recovers redshifts for objects the first pass could not measure. Where the two passes disagree, we adopt the MARZ value.

### 3.1. *Initial cross-correlation pass (RVSAO/IRAF)*

In the first pass we measured redshifts from the extracted one-dimensional spectra using the RVSAO package (Kurtz & Mink 1998) under IRAF, with the `xcsao` task performing absorption-line cross-correlation against the bundled E/S0/Sa–Sc galaxy templates and the `emsao` task centroiding the strongest emission features ([O II], H$\beta$, [O III], and H$\alpha$ where in band). Telluric bands ($\sim$ 6860, 7600 Å) were masked prior to correlation, and the Tonry & Davis (1979) $R$-value was used as the cross-correlation quality statistic. These measurements formed the basis of an earlier analysis of a subset of the present sample (Adhikari et al. 2024), but their heterogeneous, observer-dependent quality flagging motivated a uniform re-measurement.

### 3.2. *Re-measurement with* MARZ

We re-measured redshifts for all $\sim$1,400 extracted one-dimensional spectra with MARZ (Hinton et al. 2016), an interactive redshifting tool whose AUTOZ-style algorithm (Baldry et al. 2014) cross-correlates each spectrum against a library of galaxy, quasar, and stellar templates and returns a redshift together with a uniform quality

operator (QOP) on a 1–4 scale (4: certain; 3: probable; 2: tentative; 1: unusable). Using a single tool with a fixed template set and a single reviewer applying the same QOP convention to every spectrum removes the observer-to-observer inconsistencies of the first pass and yields a homogeneous quality flag that propagates cleanly into the catalog.

Two pre-processing steps were applied to the MARZ inputs. First, zero-filled detector chip gaps were detected and masked (interpolated across and assigned large variance) so that the sharp flux discontinuities at gap edges were not mis-fit as the Ca II H&K break. Second, isolated cosmic-ray and sky-line residual spikes were identified and down-weighted in the same way. Each spectrum was then inspected in MARZ and assigned a QOP.

We retain redshifts with QOP $\geq 2$, yielding 980 redshifts across the six fields (out of ∼1,400 science spectra); three of these are Galactic stars ($z \approx 0$), so 977 enter the galaxy catalog (Table 2). Of these, 912 (93%) have QOP = 4 and 44 have QOP = 3; only 24 (2%) are tentative (QOP = 2), so the dynamical results do not depend on the tentative measurements. Of these, 679 have a counterpart in the earlier IRAF pass: 628 (92%) agree to within 500 km s$^{-1}$, mutually validating the two independent measurements, and 51 are cases where the MARZ fit revises the earlier value. The remaining 302 are newly measured redshifts that the IRAF pass did not yield, predominantly from spectra the first pass had flagged as borderline. As an internal-consistency check, a second, independent MARZ pass over the same spectra reproduced the adopted redshift for >95% of objects.

### 3.3. *Redshift uncertainties*

MARZ does not return a formal per-object redshift uncertainty; the QOP flag is a discrete confidence indicator rather than a $1\sigma$ error. For the dynamical analysis that follows, cluster velocity dispersions and their uncertainties are derived directly from the member redshift distribution by biweight statistics with jackknife resampling (subsection 5.3), so they do not depend on individual $\sigma_z$ values. A formal per-object error budget, calibrated from the small number of objects observed on more than one mask, is deferred to the machine-readable catalog release; the `zerr` column is populated from the literature sources where available and left blank for MARZ measurements pending that calibration.

## 4. THE MASTER SPECTROSCOPIC CATALOG

### 4.1. *Literature sources*

We supplement our new IMACS redshifts with spectroscopic measurements drawn from four published catalogs (Bayliss et al. 2011; Carrasco et al. 2017; Sharon et al. 2020; Boschin et al. 2020) and the NASA/IPAC Extragalactic Database (NED). The breakdown of source contributions is given in Table 2.

**Table 2.** Composition of the master spectroscopic redshift catalog (post-deduplication, $0.01 < z < 7$).

| Source | $N_{\rm entries}$ | Reference |
|---|---|---|
| **IMACS (this work)** | **977** | — |
| NED | 4,952 | NASA/IPAC (NED) |
| Carrasco2017 | 958 | Carrasco et al. (2017) |
| Bayliss2011 | 251 | Bayliss et al. (2011) |
| Sharon2020 | 130 | Sharon et al. (2020) |
| Boschin2020 | 275 | Boschin et al. (2020) |
| **Total** | **7,543** | |

NOTE—Counts are post-deduplication unique-object entries with measured redshifts in the range $0.01 < z < 7$. Cross-matching used a 1.″0 tolerance; when multiple catalogs reported the same source, the smaller-uncertainty value was adopted and the alternate retained in the `z_RAW` field via the `z1;z2` convention (subsection 4.2).

### 4.2. *Cross-matching and deduplication*

Sources were cross-matched in equatorial coordinates with a 1.″0 tolerance. When the same object appeared in multiple input catalogs, we adopted the redshift with the smaller reported uncertainty and retained the alternate value(s), so that the adopted redshift is recoverable and the alternates remain available for systematic-error checks. Bad rows (no valid $z$, no valid coordinates, or $z \leq 0.01$ or $z \geq 7.0$) were removed. We did not merge our new IMACS/MARZ measurements with NED entries, since they are generally of higher quality and more uniformly characterized; rather, we flagged duplicates for the systematic source-offset checks discussed in Section 6.4.

### 4.3. *Catalog properties*

The full catalog contains 9,176 pre-deduplication entries across 74 cluster fields. After cross-matching at 1.″0 and the $0.01 < z < 7$ cut, 7,543 unique objects remain (Table 2). As summarized in subsection 2.1, 54 of the 74 fields have at least 15 cluster members and 35 have at least 30. The redshift distribution of the full catalog, color-coded by source, is shown in Figure 2.

The master catalog is available in machine-readable form in the electronic edition of this paper; columns are described in Table 3 and a representative sample is shown in Table 4.

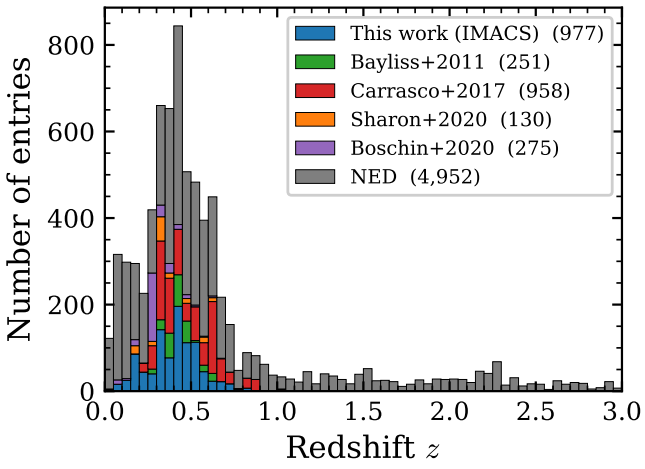



**Figure 2.** Redshift distribution of the master spectroscopic catalog, stacked by source. The new IMACS measurements (blue) preferentially target the redshift range $0.2 \lesssim z \lesssim 0.7$ where our SGAS clusters reside, while NED entries (gray) span a broader range owing to inclusion of foreground and background sources in each field.

**Table 3.** Master spectroscopic catalog columns

| Column | Description |
|---|---|
| `cluster` | SGAS cluster field name |
| `objname` | Original object designation (where available) |
| `ra_hms` | Right ascension (HH:MM:SS) |
| `dec_dms` | Declination (sDD:MM:SS) |
| `ra_deg` | RA in decimal degrees (J2000) |
| `dec_deg` | Dec in decimal degrees (J2000) |
| `z` | Adopted redshift |
| `zerr` | Redshift uncertainty |
| `source` | Provenance label (one of: `Prasanna`, `bayliss2011`, `carrasco2017`, `sharon2020`, `boschin2020`, `ned`) |
| `umag`–`zmag` | SDSS model magnitudes ($u, g, r, i, z$; AB) |
| `umag_err`–`zmag_err` | SDSS magnitude uncertainties |

We further cross-matched the catalog against SDSS DR17 photometry, attaching $u, g, r, i, z$ model magnitudes and their uncertainties to the 86% of objects with an SDSS counterpart within $1\farcs5$; the six far-southern RCS2 fields fall outside the SDSS footprint and have no photometry.

## 5. DYNAMICAL ANALYSIS PIPELINE

The complete pipeline implementation, including the helper functions described below, the seeded Monte-Carlo DS code, and the GMM substructure verification, is provided as Jupyter notebooks in the `Analysis/` directory alongside the machine-readable catalog. Re-running the notebooks reproduces the figures and the machine-readable cluster table bit-for-bit.

We process each cluster field through a five-stage pipeline: (i) velocity pre-filter to remove foreground and background sources; (ii) iterative $3\sigma$ biweight clipping to converge on the spectroscopic member catalog; (iii) biweight dynamics with jackknife uncertainties to derive $z_{\rm cl}$ and $\sigma_v$, and the dynamical mass $M_{200}$ from the $\sigma_v$–$M_{200}$ scaling of Evrard et al. (2008); (iv) the Dressler–Shectman substructure test with Monte-Carlo calibration; and (v) a Gaussian Mixture Model (GMM) in (RA, Dec, $v_{\rm pec}$) space, with the number of components $k$ selected by the Bayesian Information Criterion (BIC). Stages (iv) and (v) are independent probes of substructure and provide a cross-check on the detection.

### 5.1. *Velocity pre-filter*

For each cluster we adopt a fiducial cluster redshift $z_{\rm cl}$ from published spectroscopic biweights in the priority order Sharon et al. 2020, Bayliss et al. 2011, Carrasco et al. 2017, Boschin et al. 2020, with the literature photometric value used when no published spectroscopic determination is available. For the 9 clusters (6 of them DS-eligible) with no published cluster redshift in any of these sources, we bootstrap $z_{\rm cl}$ at runtime from a Gaussian-smoothed kernel-density estimate of the field's spectroscopic redshift distribution (KDE peak with $\sigma_{\rm kernel} = 0.02$ in $z$), followed by a biweight refinement on objects within $5{,}000\,{\rm km\,s^{-1}}$ of the peak. The KDE peak finds the dominant mode of the field distribution and is robust to contamination from lensed background sources at $z \approx 1$–$3$. The bootstrapped seed is used only to centre the velocity pre-filter; the final $z_{\rm cl}$ reported for each cluster is computed independently from the iteratively clipped member sample (subsection 5.2). Bootstrap-seeded clusters are explicitly flagged in our analysis output and are discussed collectively in subsection 8.2.

We keep as candidate cluster members only objects whose rest-frame line-of-sight velocity relative to the cluster redshift lies within $\pm 5{,}000\,{\rm km\,s^{-1}}$, i.e. $|c(z - z_{\rm cl})/(1 + z_{\rm cl})| < 5{,}000\,{\rm km\,s^{-1}}$. This window is broad enough to retain the full velocity dispersion of even the most massive systems in our sample while removing foreground galaxies and the substantial population of lensed background sources in each field. A velocity window is needed regardless of any spatial cut, because lensed

**Table 4.** Sample of the master spectroscopic catalog for the field of SDSS J2243−0935 (first 12 entries, ordered by R.A.). $u, g, r, i, z$ are SDSS model magnitudes (AB); per-band uncertainties are in the machine-readable table. The full catalog spans all 74 fields; columns are as described in Table 3.

| Object | R.A. | Decl. | $z_{\rm spec}$ | QOP | Source | $u$ | $g$ | $r$ | $i$ | $z$ |
|---|---|---|---|---|---|---|---|---|---|---|
| obj02716 | 22:42:21.840 | −09:32:08.23 | 0.1961 | 4 | This work | 22.26 | 21.23 | 20.03 | 19.55 | 19.40 |
| obj00515 | 22:42:30.240 | −09:37:58.51 | 0.1295 | 4 | This work | 23.68 | 21.91 | 19.95 | 19.14 | 18.58 |
| SDSS J224232.24-093447.2 | 22:42:32.25 | −09:34:47.2 | 2.1850 | ··· | ned | 21.74 | 21.51 | 21.00 | 20.99 | 20.60 |
| obj00428 | 22:42:32.400 | −09:27:03.17 | 0.1507 | 4 | This work | 23.67 | 21.62 | 20.39 | 19.77 | 19.30 |
| WISEA J224232.58-093955.6 | 22:42:32.58 | −09:39:55.7 | 0.9350 | ··· | ned | 20.52 | 20.40 | 20.07 | 20.27 | 19.93 |
| obj00514 | 22:42:33.538 | −09:42:53.60 | 0.4334 | 4 | This work | 23.03 | 21.78 | 20.48 | 19.87 | 19.26 |
| obj00431 | 22:42:33.600 | −09:31:44.94 | 0.5010 | 4 | This work | 24.28 | 22.67 | 21.23 | 20.48 | 19.68 |
| obj00523 | 22:42:33.655 | −09:41:19.27 | 0.5032 | 3 | This work | 25.27 | 23.58 | 21.47 | 20.92 | 20.33 |
| obj00737 | 22:42:34.560 | −09:30:06.08 | 0.5538 | 4 | This work | 22.09 | 21.47 | 19.88 | 19.07 | 18.57 |
| obj02865 | 22:42:34.800 | −09:42:00.04 | 0.2655 | 4 | This work | ··· | ··· | ··· | ··· | ··· |
| SDSS J224235.37-093427.2 | 22:42:35.37 | −09:34:27.3 | 3.5050 | ··· | ned | 25.45 | 20.97 | 20.38 | 20.07 | 19.88 |
| obj02867 | 22:42:35.427 | −09:41:26.18 | 0.1596 | 4 | This work | 22.54 | 20.86 | 20.24 | 19.96 | 19.80 |

NOTE—QOP is the MARZ quality flag (subsection 3.2); ··· where a source provides no value or has no SDSS counterpart within 1″.5. "This work" denotes our IMACS/MARZ measurements. A representative sample is shown here; the complete catalog (all fields, with photometric errors) is in the electronic edition.

sources are often spatially co-located with the cluster but kinematically unrelated. Our fiducial membership uses this velocity selection alone. Because several fields are surveyed out to 4–6 Mpc from the cluster centre, it therefore includes infall regions and structures superposed along the line of sight; we test the effect of adding a spatial aperture in subsection 6.5.

### 5.2. *Iterative biweight membership selection*

Following Beers et al. (1990), we apply iterative $3\sigma$ clipping about the biweight location $z_{\rm bi}$ and biweight scale $\sigma_z$ of the candidate-member redshifts, rejecting objects with

$$z_i < z_{\rm bi} - 3\sigma_z \quad \text{or} \quad z_i > z_{\rm bi} + 3\sigma_z, \tag{1}$$

and recomputing $z_{\rm bi}$ and $\sigma_z$ until the membership set stabilizes (usually in one or two iterations).

### 5.3. *Biweight dynamics and dynamical mass*

We adopt the biweight redshift $z_{\rm bi}$ as the cluster systemic redshift and convert the biweight scale $\sigma_z$ to a rest-frame line-of-sight velocity dispersion via

$$\sigma_v = \frac{c\,\sigma_z}{1+z_{\rm bi}}. \tag{2}$$

Uncertainties on $z_{\rm bi}$ and $\sigma_z$ are estimated by delete-one jackknife resampling, and we report 95% confidence intervals as $z_{\rm bi} \pm 1.96\,\sigma_{\rm JK}$, where $\sigma_{\rm JK}$ is the jackknife standard error.

Dynamical masses $M_{200}$ are computed from $\sigma_v$ using the $N$-body calibration of Evrard et al. (2008):

$$\sigma_v = \sigma_{15}\left[\frac{h(z)\,M_{200}}{10^{15} M_\odot}\right]^{\alpha}, \tag{3}$$

with $\sigma_{15} = 1082.9\,{\rm km\,s^{-1}}$, $\alpha = 0.3361$, and $h(z) = H(z)/100\,{\rm km\,s^{-1}\,Mpc^{-1}}$. Because this relation assumes virial equilibrium, $M_{200}$ is an upper limit for the substructured clusters (Section 6), whose $\sigma_v$ is inflated by unrelaxed motions.

### 5.4. *Dressler–Shectman test*

For each member galaxy $i$ we compute the local mean velocity $v_{{\rm local},i}$ and dispersion $\sigma_{{\rm local},i}$ from the $N_{\rm nn}$ nearest neighbors in projected sky position, where $N_{\rm nn} = \max(8, \lfloor\sqrt{N}\rfloor)$, i.e. $\lfloor\sqrt{N}\rfloor$ following Pinkney et al. (1996) with a floor of eight neighbors for the smaller clusters. The local-to-global deviation is

$$\delta_i^2 = \frac{N_{\rm nn}+1}{\sigma_v^2}\left[(v_{{\rm local},i} - v_{\rm cl})^2 + (\sigma_{{\rm local},i} - \sigma_v)^2\right], \tag{4}$$

and the cluster-level statistic is $\Delta_{\rm DS} = \sum_i \delta_i$. We assess the significance of $\Delta_{\rm DS}$ against a null distribution built from $10^3$ Monte-Carlo realizations in which the member velocities are randomly shuffled among the member sky positions, following Dressler & Shectman (1988), so that any real position–velocity correlation is destroyed while the two distributions are individually preserved. The shuffles use a fixed random seed so the reported $p$-values are exactly reproducible. The significance is

$$p_{\rm DS} = \frac{1}{N_{\rm MC}}\sum_{k=1}^{N_{\rm MC}} \Theta\!\left(\Delta_{\rm DS}^{(k)} - \Delta_{\rm DS}^{\rm obs}\right), \tag{5}$$

where $\Delta_{\rm DS}^{(k)}$ is the statistic of the $k$-th shuffle, $\Delta_{\rm DS}^{\rm obs}$ the observed value, $N_{\rm MC} = 10^3$, and $\Theta(x) = 1$ for $x \geq 0$ and 0 otherwise. A cluster with no shuffle reaching the observed value is reported as $p_{\rm DS} < 0.001$.

Bubble plots showing the spatial distribution of $\delta_i$ are constructed by plotting each member at its sky position with a marker area proportional to $e^{\delta_i}$, the standard convention (Dressler & Shectman 1988); clumps of inflated bubbles indicate candidate subcluster cores.

### 5.5. *GMM substructure verification*

The DS test produces a global $p$-value and a per-galaxy deviation $\delta_i$, but does not directly specify how many subclusters are present or which members are assigned to each. As an independent parametric check, we fit a Gaussian Mixture Model (GMM) in the standardized (RA, Dec, $v_{\rm pec}$) space using the SCIKIT-LEARN implementation (Pedregosa et al. 2011), with full covariance matrices and 10 random initializations per fit. We fit models for $k = 1, 2, 3, 4$ components and select the BIC-preferred $k$, requiring $\Delta\mathrm{BIC} \geq 6$ over $k-1$ for "decisive" evidence (Kass & Raftery 1995). We report the per-component biweight redshift, mean peculiar velocity, and velocity dispersion, and use the hard GMM assignments to construct sky maps showing the spatial coherence (or lack thereof) of each kinematic component.

A robust substructure detection in this paper requires both $p_{\rm DS} < 0.01$ and a BIC-preferred $k \geq 2$ with $\Delta\mathrm{BIC}(k_{\rm best} - 1 \to k_{\rm best}) \geq 6$.

## 6. RESULTS

### 6.1. *Sample-wide dynamical properties*

Table 5 summarizes our dynamical results for the 54 clusters meeting the $N \geq 15$ dynamics threshold. We list the number of confirmed members after iterative biweight convergence ($N_{\rm mem}$), biweight redshift ($z_{\rm bi}$), rest-frame line-of-sight velocity dispersion ($\sigma_v$), dynamical mass ($M_{200}$), observed $\Delta_{\rm DS}/N$, the Monte-Carlo $p$-value, and our substructure flag. The DS test loses statistical power below $N_{\rm mem} = 30$ (Pinkney et al. 1996, see subsection 5.4), so for the 11 clusters with $20 \leq N_{\rm mem} < 30$ the DS values are listed for completeness but marked with a star and excluded from the substructure statistics; they enter only the $N \geq 20$ X-ray comparison of section 7. For the 8 clusters with $15 \leq N_{\rm mem} < 20$ the DS test was not run.

The $\sigma_v$ distribution of our sample has median $971\,\mathrm{km\,s^{-1}}$, with 15 of 54 clusters above $1{,}200\,\mathrm{km\,s^{-1}}$ and a tail extending to $\sigma_v \sim 2{,}500\,\mathrm{km\,s^{-1}}$ (Figure 3). The dynamical-mass distribution spans $M_{200} \sim 0.1$–$90 \times 10^{14} M_\odot$, but the extreme upper tail is not reliable: the largest dispersions (e.g. SDSS J1055+5547, SDSS J1050+0017, and SDSS J1336−0331) are inflated by galaxies beyond $R_{200}$ or by superposed groups, and fall to $\sim$500–800 $\mathrm{km\,s^{-1}}$ when membership is restricted to the cluster (subsection 6.5).

### 6.2. *Substructure detections*

We detect kinematic substructure ($p_{\rm DS} \leq 0.05$, Equation 5) in 13 of the 35 DS-eligible clusters, yielding a kinematic-substructure fraction of $f_{\rm sub} = 0.37 \pm 0.08$ (binomial $1\sigma$) for the DS-eligible subsample. Adopting a more conservative threshold ($p_{\rm DS} < 0.01$) instead flags 6 clusters ($f_{\rm sub} = 0.17 \pm 0.06$).

The five most significant detections all have $p_{\rm DS} < 0.001$ (0 of 1000 Monte-Carlo realisations as extreme as observed): SDSS J2243−0935, SDSS J1038+4849, SDSS J1050+0017, SDSS J1152+0930, and SDSS J1439+1208. The remaining eight detections span $0.007 \leq p_{\rm DS} \leq 0.036$ (RCS2 J2329−1317, SDSS J0915+3826, SDSS J0146−0929, RCS2 J1125−0628, SDSS J0851+3331, SDSS J1336−0331, SDSS J2111−0114, and SDSS J1152+3313); all thirteen are listed with their statistics in Table 5. Of these, SDSS J2243−0935 provides the clearest pedagogical example of a near-line-of-sight major merger (subsection 6.3). Only three of the thirteen detections (SDSS J1152+0930, RCS2 J2329−1317, and SDSS J2111−0114) remain significant in both of the cluster apertures tested in subsection 6.5.

### 6.3. *Case study: SDSS J2243−0935*

SDSS J2243−0935 provides the clearest and most pedagogically illustrative substructure detection in our sample, and we discuss it in detail. The cluster lies at $z_{\rm cl} = 0.4468$ and has $N_{\rm mem} = 134$ confirmed spectroscopic members. The biweight statistics yield a line-of-sight velocity dispersion of $\sigma_v = 1358 \pm 88\,\mathrm{km\,s^{-1}}$ and a dynamical mass of $M_{200} = 1.54 \times 10^{15} M_\odot$. These values place SDSS J2243−0935 among the most massive clusters in our sample, but as we now show, the high $\sigma_v$ is significantly inflated by ongoing merger activity.

*DS detection.*—The seeded DS test gives $\Delta_{\rm DS}/N = 1.94$ against a Monte-Carlo null distribution centred at 1.13 with $p_{\rm DS} < 0.001$ (0 of 1000 paired-permutation realisations produced a value at least as extreme; Figure 5a). The per-galaxy $\delta_i$ statistics flag a clear top-15 high-$\delta_i$ population, which divides cleanly into two spatial clumps with opposite velocity offsets: a NE clump redshifted by $\sim +2000\,\mathrm{km\,s^{-1}}$ and a SW clump blueshifted by $\sim -2700\,\mathrm{km\,s^{-1}}$, a relative line-of-sight velocity of $\sim 5000\,\mathrm{km\,s^{-1}}$ characteristic of a near-line-of-sight major merger. These two clumps are visible as the flanking

**Table 5.** Dynamical and substructure properties of the $N_{\rm mem} \geq 15$ cluster sample, sorted by $p_{\rm DS}$ ascending so the most significant detections appear first (Part 1 of 2).

| Cluster | $N_{\rm mem}$ | $z_{\rm bi}$ | $\sigma_v$ (km s$^{-1}$) | $M_{200}$ ($10^{14}\,M_\odot$) | $\Delta_{\rm DS}/N$ | $p_{\rm DS}$ | Sub. flag |
|---|---|---|---|---|---|---|---|
| SDSS J1050+0017 | 107 | 0.5951 | $1860 \pm 302$ | 35.83 | 2.74 | $<0.001$ | Y |
| SDSS J1439+1208 | 69 | 0.4324 | $1833 \pm 175$ | 37.87 | 1.89 | $<0.001$ | Y |
| SDSS J1152+0930 | 88 | 0.5133 | $1452 \pm 205$ | 18.04 | 2.08 | $<0.001$ | Y |
| SDSS J1038+4849 | 128 | 0.4324 | $1363 \pm 113$ | 15.68 | 2.11 | $<0.001$ | Y |
| SDSS J2243−0935 | 134 | 0.4468 | $1358 \pm 88$ | 15.36 | 1.94 | $<0.001$ | Y |
| RCS2 J2329−1317 | 79 | 0.3908 | $1033 \pm 98$ | 7.04 | 1.57 | 0.007 | Y |
| SDSS J0915+3826 | 72 | 0.3957 | $859 \pm 94$ | 4.06 | 1.58 | 0.011 | Y |
| SDSS J0146−0929 | 65 | 0.4471 | $1593 \pm 165$ | 24.71 | 1.53 | 0.012 | Y |
| RCS2 J1125−0628 | 33 | 0.4745 | $745 \pm 126$ | 2.54 | 1.60 | 0.012 | Y |
| SDSS J1336−0331 | 66 | 0.1757 | $1934 \pm 536$ | 51.46 | 1.99 | 0.015 | Y |
| SDSS J0851+3331 | 67 | 0.3688 | $670 \pm 62$ | 1.97 | 1.42 | 0.015 | Y |
| SDSS J2111−0114 | 52 | 0.6359 | $1097 \pm 120$ | 7.27 | 1.45 | 0.023 | Y |
| SDSS J1152+3313 | 54 | 0.3615 | $685 \pm 83$ | 2.11 | 1.43 | 0.036 | Y |
| Abell 1703 | 152 | 0.2769 | $1367 \pm 78$ | 17.32 | 1.23 | 0.059 | N |
| SDSS J1055+5547 | 33 | 0.4656 | $2505 \pm 1043$ | 93.98 | 1.75 | 0.067 | N |
| RCS2 J1055−0459 | 38 | 0.6076 | $730 \pm 75$ | 2.20 | 1.24 | 0.073 | N |
| SDSS J1329+2243 | 57 | 0.4434 | $655 \pm 103$ | 1.76 | 1.61 | 0.092 | N |
| RCS2 J2336−0608 | 34 | 0.3923 | $1016 \pm 120$ | 6.71 | 1.21 | 0.152 | N |
| SDSS J1138+2754 | 36 | 0.4485 | $1377 \pm 317$ | 16.00 | 1.54 | 0.221 | N |
| SDSS J1420+3955 | 37 | 0.6067 | $1195 \pm 134$ | 9.54 | 1.10 | 0.266 | N |
| RCS2 J2151−0138 | 49 | 0.3160 | $1036 \pm 112$ | 7.41 | 1.06 | 0.324 | N |
| SDSS J1621+0607 | 93 | 0.3429 | $1078 \pm 143$ | 8.22 | 1.50 | 0.363 | N |
| RCS2 J2329−0120 | 39 | 0.5312 | $1013 \pm 106$ | 6.12 | 0.95 | 0.452 | N |
| SDSS J1002+2031 | 50 | 0.3190 | $1146 \pm 132$ | 9.99 | 1.12 | 0.485 | N |
| RCS2 J1250+0244 | 32 | 0.6904 | $648 \pm 90$ | 1.47 | 1.04 | 0.575 | N |
| SDSS J0928+2031 | 32 | 0.1917 | $818 \pm 135$ | 3.94 | 0.99 | 0.583 | N |
| RCS2 J0047+0508 | 49 | 0.4287 | $745 \pm 92$ | 2.60 | 1.08 | 0.593 | N |

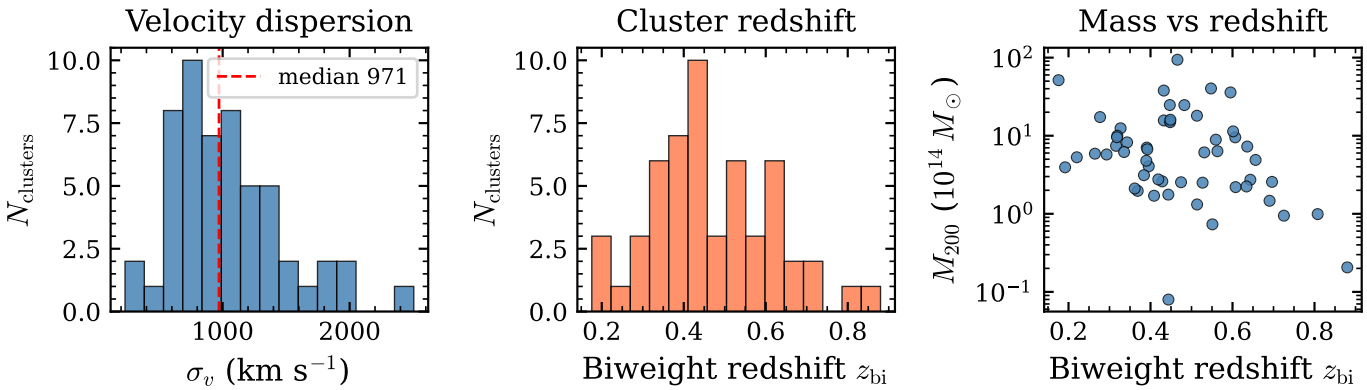



**Figure 3.** Sample-wide dynamical properties of the 54 $N \geq 15$ clusters. *Left:* velocity dispersion distribution (median 971 km s$^{-1}$, dashed red). *Centre:* biweight redshift distribution. *Right:* dynamical mass $M_{200}$ against biweight redshift, showing how the sample populates the mass–redshift plane.

groups of inflated bubbles in the DS bubble plot (Figure 5b) and are listed in Table 7.

*GMM verification.*—As an independent parametric check we fit Gaussian Mixture Models for $k = 1, \ldots, 4$ in the standardized (RA, Dec, $v_{\rm pec}$) space. The BIC scan

**Table 6.** Dynamical and substructure properties of the $N_{\rm mem} \geq 15$ cluster sample (Part 2 of 2; continued from Table 5).

| Cluster | $N_{\rm mem}$ | $z_{\rm bi}$ | $\sigma_v$ | $M_{200}$ | $\Delta_{\rm DS}/N$ | $p_{\rm DS}$ | Sub. |
|---|---|---|---|---|---|---|---|
| | | | (km s$^{-1}$) | ($10^{14}\,M_\odot$) | | | flag |
| SDSS J0957+0509 | 39 | 0.4479 | $1344 \pm 215$ | 14.91 | 1.18 | 0.620 | N |
| SDSS J1519+0840 | 46 | 0.3182 | $1129 \pm 97$ | 9.58 | 0.81 | 0.707 | N |
| RCS2 J0327−1326 | 44 | 0.5632 | $1032 \pm 116$ | 6.33 | 0.91 | 0.791 | N |
| RCS2 J1108−0456 | 46 | 0.4086 | $643 \pm 63$ | 1.70 | 0.80 | 0.818 | N |
| RCS2 J0252−1459 | 36 | 0.2644 | $949 \pm 114$ | 5.88 | 0.81 | 0.831 | N |
| SDSS J1209+2640 | 56 | 0.5593 | $1156 \pm 152$ | 8.89 | 1.05 | 0.858 | N |
| SDSS J2135−0102 | 86 | 0.3270 | $1235 \pm 93$ | 12.45 | 0.78 | 0.978 | N |
| SDSS J1517+1003 | 57 | 0.6434 | $790 \pm 88$ | 2.73 | 0.73 | 0.995 | N |
| RCS2 J2147−0102 | 25 | 0.8799 | $348 \pm 58$ | 0.21 | 1.32 | 0.070$^\star$ | N$^\star$ |
| SDSS J1527+0652 | 29 | 0.3897 | $906 \pm 114$ | 4.78 | 1.11 | 0.185$^\star$ | N$^\star$ |
| SDSS J1111+1408 | 20 | 0.2202 | $908 \pm 124$ | 5.30 | 0.98 | 0.198$^\star$ | N$^\star$ |
| RCS2 J1526+0432 | 29 | 0.6341 | $738 \pm 81$ | 2.24 | 1.09 | 0.247$^\star$ | N$^\star$ |
| RCS2 J0052+0433 | 23 | 0.7254 | $563 \pm 148$ | 0.95 | 1.40 | 0.258$^\star$ | N$^\star$ |
| SDSS J1456+5702 | 25 | 0.4825 | $1603 \pm 307$ | 24.67 | 1.23 | 0.379$^\star$ | N$^\star$ |
| RCS2 J0034+0225 | 24 | 0.3839 | $785 \pm 192$ | 3.12 | 1.11 | 0.471$^\star$ | N$^\star$ |
| RCS2 J0309−1437 | 23 | 0.8078 | $582 \pm 56$ | 0.99 | 0.60 | 0.779$^\star$ | N$^\star$ |
| SDSS J0108+0624 | 27 | 0.5475 | $1916 \pm 835$ | 40.29 | 1.83 | 0.780$^\star$ | N$^\star$ |
| SDSS J2313−0104 | 23 | 0.5271 | $750 \pm 114$ | 2.50 | 0.68 | 0.818$^\star$ | N$^\star$ |
| SDSS J1110+6459 | 21 | 0.6563 | $965 \pm 174$ | 4.90 | 0.73 | 0.842$^\star$ | N$^\star$ |
| SDSS J1522+2535 | 19 | 0.6015 | $1266 \pm 167$ | 11.37 | ··· | ··· | ··· |
| SDSS J1531+3414 | 19 | 0.3355 | $978 \pm 153$ | 6.19 | ··· | ··· | ··· |
| RCS2 J0057+0209 | 19 | 0.2929 | $947 \pm 144$ | 5.75 | ··· | ··· | ··· |
| RCS2 J0038+0215 | 19 | 0.6964 | $782 \pm 104$ | 2.56 | ··· | ··· | ··· |
| SDSS J1343+4155 | 15 | 0.4190 | $758 \pm 208$ | 2.76 | ··· | ··· | ··· |
| SDSS J0004−0103 | 19 | 0.5136 | $603 \pm 91$ | 1.32 | ··· | ··· | ··· |
| RCS2 J1511+0630 | 15 | 0.5510 | $499 \pm 121$ | 0.73 | ··· | ··· | ··· |
| SDSS J1723+3411 | 15 | 0.4437 | $232 \pm 45$ | 0.08 | ··· | ··· | ··· |

NOTE—$N_{\rm mem}$: spectroscopic members after iterative $3\sigma$ biweight convergence. $z_{\rm bi}$: biweight cluster redshift. $\sigma_v$: rest-frame line-of-sight velocity dispersion, error from delete-one jackknife. $M_{200}$: dynamical mass from the $\sigma_v$ scaling of Evrard et al. (2008); treat as an upper limit when the substructure flag is Y (see subsection 5.3). $\Delta_{\rm DS}/N$: Dressler–Shectman statistic; expected $\approx 1$ under no-substructure null. $p_{\rm DS}$: fraction of $10^3$ Monte-Carlo permutations with $\Delta_{\rm DS}/N \geq$ observed. Sub. flag: Y if $p_{\rm DS} \leq 0.05$, N if $p_{\rm DS} > 0.05$. $^\star$: $20 \leq N_{\rm mem} < 30$, below the $N \geq 30$ threshold for a reliable DS test (Pinkney et al. 1996); listed for completeness and excluded from the substructure statistics. ··· : DS test not run ($N_{\rm mem} < 20$).

yields 1152.6, 1129.9, 1132.2, and 1118.8, respectively. $k = 2$ is decisively preferred over $k = 1$ ($\Delta$BIC = 22.7), whereas $k = 3$ does *not* improve on $k = 2$ ($\Delta$BIC = −2.4); we therefore adopt the $k = 2$ decomposition as our fiducial (the marginally lower BIC at $k = 4$ is consistent with over-fitting the high-$\delta_i$ tail and is not decisive). The two components have 65 and 69 members, mean peculiar velocities of +688 and −690 km s$^{-1}$, velocity dispersions of 1095 and 1253 km s$^{-1}$, and biweight redshifts of 0.4498 and 0.4438, respectively, separated by $\sim 1{,}380$ km s$^{-1}$ in mean velocity and offset on the sky into a NE and a SW grouping (Figure 5c).

Crucially, the redshifted GMM component ($\langle v_{\rm pec} \rangle \approx +690$ km s$^{-1}$) is spatially coincident with the NE clump of high-$\delta_i$ galaxies in the DS bubble plot (Figure 5b), while the blueshifted component occupies the SW field where the most extreme negative-$v_{\rm pec}$ high-$\delta_i$ galaxies sit (Table 7). Two statistically independent tests, the non-parametric DS statistic and the parametric GMM, thus localise the same bimodal kinematic structure, which is the strongest evidence for the ongoing-merger interpretation.

**Table 7.** Top-15 high-$\delta_i$ candidate substructure members in SDSS J2243−0935.

| RA (deg) | Dec (deg) | $z$ | $v_{\rm pec}$ (km s$^{-1}$) | $\delta_i$ |
|---|---|---|---|---|
| 340.8690 | −9.4674 | 0.4561 | +1934 | 4.56 |
| 340.6528 | −9.7543 | 0.4351 | −2434 | 4.46 |
| 340.6769 | −9.7641 | 0.4340 | −2662 | 4.46 |
| 340.6397 | −9.7149 | 0.4334 | −2770 | 4.38 |
| 340.6747 | −9.6875 | 0.4341 | −2631 | 4.38 |
| 340.6523 | −9.7080 | 0.4479 | +222 | 4.38 |
| 340.8630 | −9.4665 | 0.4562 | +1942 | 4.12 |
| 340.8535 | −9.4407 | 0.4483 | +315 | 4.06 |
| 340.8770 | −9.4721 | 0.4588 | +2476 | 3.95 |
| 340.8809 | −9.4808 | 0.4514 | +943 | 3.95 |
| 340.8710 | −9.4725 | 0.4544 | +1579 | 3.95 |
| 340.8800 | −9.4634 | 0.4569 | +2085 | 3.95 |
| 340.8880 | −9.4791 | 0.4549 | +1675 | 3.77 |
| 340.8877 | −9.4887 | 0.4575 | +2220 | 3.46 |
| 340.9210 | −9.6684 | 0.4490 | +456 | 3.39 |

NOTE—Members are sorted by descending $\delta_i$. The high-$\delta_i$ population splits cleanly into two spatial clumps with opposite velocity offsets: a NE clump around $(\alpha, \delta) \approx (340.^\circ87, -9.^\circ47)$ redshifted by $\sim+2000$ km s$^{-1}$ and a SW clump around $(340.^\circ65, -9.^\circ72)$ blueshifted by $\sim-2700$ km s$^{-1}$. The relative line-of-sight velocity of $\sim$ 5000 km s$^{-1}$ is characteristic of an ongoing major merger viewed near-line-of-sight (subsection 6.3).

*Inferred merger geometry.*—The combined DS+GMM analysis indicates a near-line-of-sight major merger. The two bulk GMM components are separated by $\sim$ 1,380 km s$^{-1}$ in mean velocity, while the high-$\delta_i$ substructure cores span a wider $\sim$ 5000 km s$^{-1}$ in line-of-sight velocity (Table 7). This is the signature of a merger axis oriented close to the line of sight, in which the largest velocity offsets are carried by the densest subclumps. The high $\sigma_v = 1358$ km s$^{-1}$ derived for the cluster as a whole is correspondingly inflated by this kinematic superposition; the dispersions of the individual GMM components are $\sim$ 1,100–1,250 km s$^{-1}$, suggesting that the constituent halos are individually less massive than the $\sigma_v$-inferred $M_{200}$ for the system taken as a single virialized cluster.

Figures 4 and 5 summarize the analysis for this cluster. Figure 4 shows the bulk dynamical evidence (the phase-space diagram and the member redshift distribution), while Figure 5 establishes the substructure through the DS Monte-Carlo significance, the DS bubble plot, and the $k = 2$ GMM decomposition.

*Dependence on aperture.*—The DS significance of SDSS J2243−0935 depends on members at large radius: 54% of its members lie beyond the iterated $R_{200} = 2.0$ Mpc, and restricting membership to $R < R_{200}$ ($N = 62$, $\sigma_v = 1329$ km s$^{-1}$) or $R < 1.5$ Mpc ($N = 48$) gives $p_{\rm DS} = 0.16$ and 0.27, respectively (subsection 6.5). The X-ray evidence for a merger in this system (subsection 7.4) is unaffected, but the kinematic detection should be read as reflecting structure in the cluster outskirts as well as in the core.

### 6.4. *Source-catalog robustness check*

Because the master catalog merges several input catalogs (Table 2), a natural concern is that a systematic redshift zeropoint offset between two source catalogs could mimic a bimodal velocity structure and produce a spurious DS detection. Two features of the analysis bound this risk. First, the most significant detections are dominated by a single source label: the member samples of the highest-significance clusters (e.g. SDSS J2243−0935, SDSS J1038+4849, SDSS J1050+0017) are drawn overwhelmingly from a single homogeneous source (either our MARZ measurements or a single literature catalog), so the detected velocity structure cannot arise from a cross-catalog zeropoint. Second, our merge does not blend redshifts: when an object appears in multiple catalogs we adopt one value and retain the alternates separately (subsection 4.2), so no member's velocity is a catalog-dependent average. We have re-run the DS test on the largest single-source subset (our IMACS measurements or NED) of each of the seven X-ray-overlap clusters with DS detections (section 7). Six remain significant ($p_{\rm DS} \leq 0.015$); SDSS J0851+3331 becomes marginal ($p_{\rm DS} = 0.055$ for its 54 NED members). Source mixing therefore does not drive these detections. The catalog provides the source label on every member so that this check can be reproduced for the full sample.

### 6.5. *Dependence on the membership aperture*

The X-ray morphological parameters we compare against in section 7 describe the ICM of the cluster itself, whereas our velocity-only membership extends to 4–6 Mpc in the most widely surveyed fields. In SDSS J1050+0017, SDSS J1439+1208, and SDSS J1038+4849, for example, 84%, 72%, and 62% of the members lie beyond the iterated $R_{200}$ defined below. To test whether the DS results describe the clusters themselves, we repeated the full analysis with two spatial apertures applied before the $3\sigma$ biweight clipping: (i) an iterated $R_{200}$, in which we start from $R < 1.5$ Mpc, clip, compute $\sigma_v$ and $M_{200}$ (subsection 5.3), set $R_{200} = [3M_{200}/(4\pi\, 200\, \rho_{\rm c}(z))]^{1/3}$, and repeat until $R_{200}$ changes by less than 0.01 Mpc (each step is limited to a factor of 1.5, and where the iteration settles into a two-cycle, as for 10 of the 54 clusters, we adopt the smaller radius);

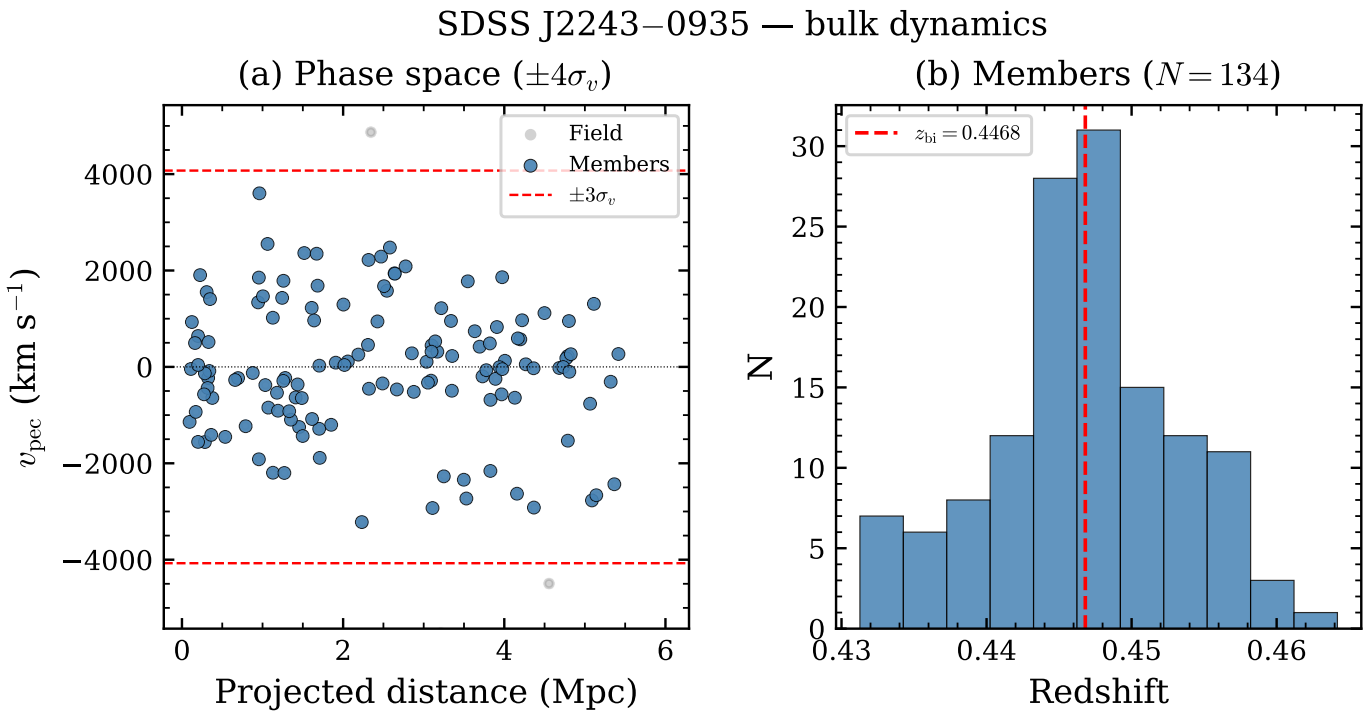



**Figure 4.** Bulk dynamics of SDSS J2243−0935. (a) Phase-space (projected distance vs. $v_{\rm pec}$) diagram zoomed to $\pm 4\sigma_v$ so the $\pm 3\sigma_v$ member envelope is visible. (b) Redshift histogram of the $N_{\rm mem} = 134$ members with the biweight $z_{\rm bi}$ marked.

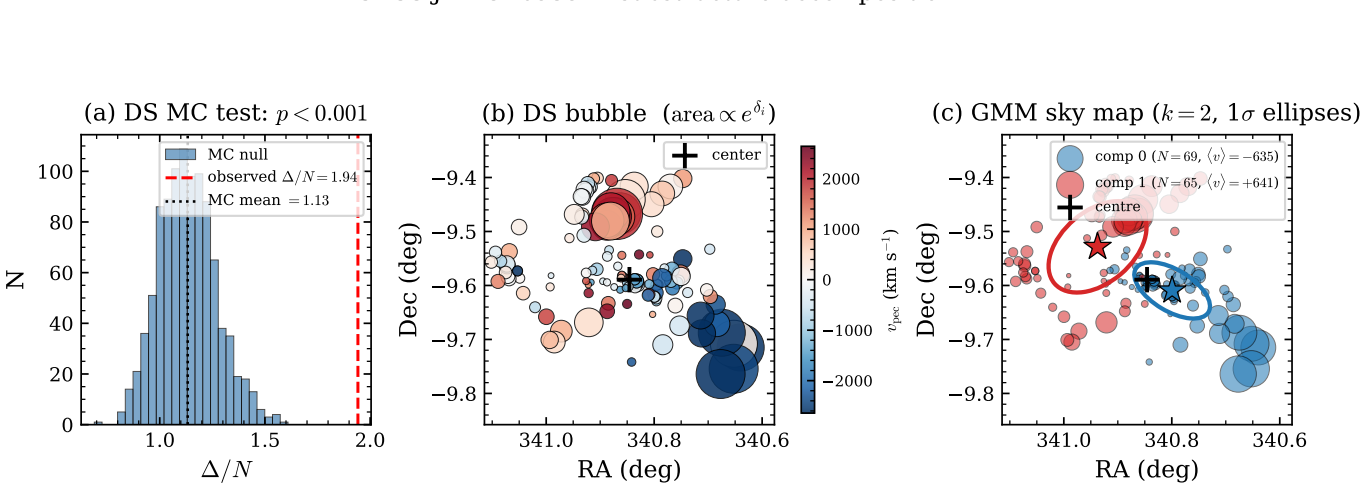



**Figure 5.** Substructure decomposition of SDSS J2243−0935. (a) Seeded DS Monte-Carlo null distribution; the red dashed line is the observed $\Delta_{\rm DS}/N$ (far in the tail, $p_{\rm DS} < 0.001$) and the dotted line the MC mean. (b) DS bubble plot, marker area $\propto e^{\delta_i}$ and colour-coded by peculiar velocity; two spatially coherent clumps of inflated bubbles flank the cluster centre. (c) GMM sky map for the BIC-preferred $k = 2$ decomposition, with $1\sigma$ covariance ellipses and starred component centroids.

and (ii) a fixed $R < 1.5$ Mpc, which does not depend on $\sigma_v$. All other settings are unchanged. The results are summarized in Table 8.

*Substructure fraction.*—The kinematic substructure fraction falls from $0.37 \pm 0.08$ with velocity-only membership to $0.15 \pm 0.07$ within $R_{200}$ and $0.11 \pm 0.06$ within 1.5 Mpc. The detections that survive both apertures are SDSS J1152+0930, RCS2 J2329−1317, and SDSS J2111−0114 (RCS2 J1125−0628 also survives within $R_{200}$, $p_{\rm DS} = 0.025$). In RCS2 J2329−1317, SDSS J2111−0114, and RCS2 J1125−0628, at most 12% of the velocity-only members lie beyond $R_{200}$, so their detections never depended on the outskirts. Every detection that disappears comes from a field whose mem-

**Table 8.** Kinematic substructure and agreement with the X-ray classification as a function of the membership aperture.

| Membership | $N_{\rm DS}$ ($N \geq 30$) | $f_{\rm sub}$ ($N \geq 30$) | Agreement ($N \geq 20$) | Agreement ($N \geq 30$) | Kin. disturbed, X-ray relaxed | Kin. relaxed, X-ray disturbed |
|---|---|---|---|---|---|---|
| Velocity only (fiducial) | 35 | $13/35 = 0.37 \pm 0.08$ | 5/18 (28%) | 5/16 (31%) | 5 | 8 |
| $R < R_{200}$ (iterated) | 27 | $4/27 = 0.15 \pm 0.07$ | 5/16 (31%) | 3/10 (30%) | 1 | 10 |
| $R < 1.5\,$Mpc | 27 | $3/27 = 0.11 \pm 0.06$ | 7/17 (41%) | 5/10 (50%) | 1 | 9 |

NOTE—$N_{\rm DS}$ is the number of clusters with $N \geq 30$ members inside the aperture and $f_{\rm sub}$ the fraction with $p_{\rm DS} \leq 0.05$ (binomial $1\sigma$). Agreement is between the binary kinematic and X-ray classes (section 7) for the overlap clusters with at least the stated number of members inside the aperture. The last two columns count the off-diagonal cells for the $N \geq 20$ overlap sample.

bers extend to at least 3 Mpc from the cluster centre. Several of the lost detections trace identifiable structures. The velocity distribution of SDSS J1050+0017 is decisively two-component ($\Delta$BIC = 64 for a one-dimensional GMM in $v_{\rm pec}$): 21% of its velocity-only members form a group at $+4050\,{\rm km\,s^{-1}}$ with an internal dispersion of $\sim 360\,{\rm km\,s^{-1}}$, located mostly 1.5–5.5 Mpc from the cluster centre, and within 1.5 Mpc the cluster has $\sigma_v = 571\,{\rm km\,s^{-1}}$ and $p_{\rm DS} = 0.24$. SDSS J1336−0331 contains a group of about seven galaxies near $-3700\,{\rm km\,s^{-1}}$ within 1.5 Mpc, and its clipped membership is bistable, converging either on a cold core ($\sigma_v \approx 500\,{\rm km\,s^{-1}}$ within 1.5 Mpc, $p_{\rm DS} = 0.88$) or on core plus group ($\sigma_v \approx 1930\,{\rm km\,s^{-1}}$); its velocity-only detection also disappears ($p_{\rm DS} = 0.19$) when the three highest-$\delta_i$ galaxies are removed. In SDSS J1038+4849 and SDSS J1439+1208 the signal comes from the outskirts ($p_{\rm DS} = 0.19$ within $R_{200}$ and 0.90 within 1.5 Mpc, respectively), and the $\sigma_v = 2505\,{\rm km\,s^{-1}}$ of SDSS J1055+5547 falls to $720\,{\rm km\,s^{-1}}$ within $R_{200}$.

*SDSS J1152+0930.*—This cluster is the exception. It has $p_{\rm DS} < 0.001$ in both apertures ($N = 41$ within $R_{200} = 2.07\,$Mpc and $N = 28$ within 1.5 Mpc), and the detection survives removing the five highest-$\delta_i$ galaxies, removing members beyond $2\sigma_v$, restricting to our IMACS redshifts alone, and every delete-one jackknife subsample. Its member velocity distribution is non-Gaussian (Shapiro–Wilk $p = 0.002$), with blueshifted and redshifted members segregated on opposite sides of the core. Since Gassis et al. (2026) classify it as moderately relaxed ($M = 0.29$), it is the strongest candidate in our sample for a merger oriented close to the line of sight.

*Detection power.*—Restricting membership lowers $N$, so a lower detection rate could simply reflect lost sensitivity. We tested this with mock mergers placed on the observed sky positions of the 16 X-ray-overlap clusters that retain $N \geq 20$ members within $R_{200}$. For each cluster we assigned the velocity-only members Gaussian velocities with dispersion equal to the in-aperture $\sigma_v$. In merger mocks, a compact subgroup made of the $fN_{200}$ galaxies inside $R_{200}$ nearest a randomly chosen galaxy inside $R_{200}$ is offset along the line of sight by $\Delta v$ (random sign) with an internal dispersion of $0.5\sigma_v$, for $f = 0.15$, 0.25, and 0.40 and $\Delta v = 0.5$–$2\,\sigma_v$. We ran the DS test (300 shuffles) on 200 mocks per configuration, both on the full velocity-only member set and on the members inside $R_{200}$, so that the same core merger is seen through both apertures. The false-positive rate on relaxed mocks is 5% in both cases (2–9% per cluster), as expected. A core merger with $f = 0.25$ and $\Delta v = 1.5\sigma_v$ is detected at least as often inside $R_{200}$ as with velocity-only membership (mean 60% versus 56%; Figure 6), because the additional outskirts galaxies dilute the local signal. The power is modest for the smallest clusters (24–36% at $N_{200} \approx 20$–26). The fall in $f_{\rm sub}$ is therefore not a loss of sensitivity: the velocity-only detections that disappear were produced by structure outside $R_{200}$.

The same mocks constrain the X-ray-disturbed clusters. None of the 10 X-ray-disturbed clusters in this set shows DS substructure within $R_{200}$. Had each hosted a subcluster with $f = 0.25$ offset by $\Delta v = 1\sigma_v$ ($1.5\sigma_v$), we would expect 3.3 (5.6) detections, with a probability of observing none of 0.01 ($2 \times 10^{-5}$); only small offsets ($\Delta v = 0.5\sigma_v$, 1.6 expected, probability 0.18) are consistent with the data. These mocks are idealized (Gaussian velocities, compact subgroups, no infall), but the conclusion that the X-ray-disturbed clusters lack large line-of-sight velocity substructure in their cores does not depend on the details.

## 7. COMPARISON WITH X-RAY DYNAMICAL STATE

The Chandra Strong Lens Sample of Gassis et al. (2026) provides X-ray morphological dynamical-state proxies for a sample of 28 strong-lensing clusters that overlaps our spectroscopic sample. Gassis et al. (2026) measure four morphological parameters from Chandra ACIS-I/S imaging (the concentration parameter $c$, the asymmetry parameter $A$, the centroid shift $\log(w)$, and the X-ray–BCG centroid separation $D$ in kpc) and combine them into a single relaxation parameter $M$ follow-

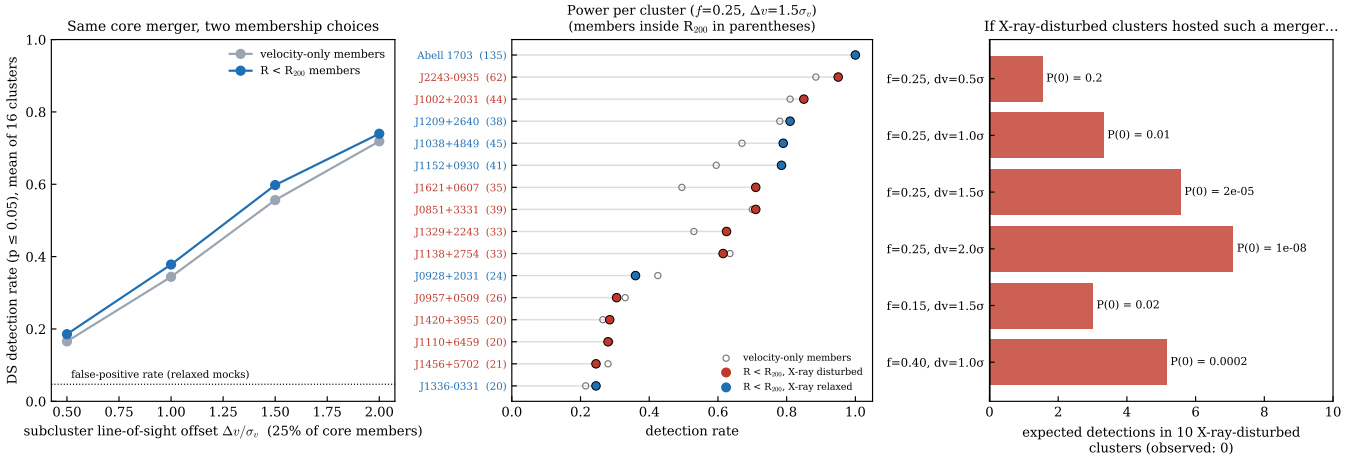



**Figure 6.** DS detection power from mock mergers on the observed galaxy positions of the 16 X-ray-overlap clusters with $N \geq 20$ members inside $R_{200}$. *Left:* mean detection rate ($p_{\rm DS} \leq 0.05$) for a subcluster containing 25% of the core members, as a function of its line-of-sight offset, for the velocity-only member set (gray) and for the members inside $R_{200}$ (blue); the dotted line is the false-positive rate for relaxed mocks. *Centre:* per-cluster detection rate for $f = 0.25$, $\Delta v = 1.5\sigma_v$, sorted from highest to lowest; filled circles use the members inside $R_{200}$ (number in parentheses) and are colored by X-ray class (red disturbed, blue relaxed), and open gray circles use the velocity-only member set. *Right:* expected number of DS detections among the 10 X-ray-disturbed clusters if each hosted the indicated merger, with the probability of observing none; none is observed.

ing Rasia et al. (2013), classifying each cluster as relaxed, moderately relaxed (M–relaxed), moderately disturbed (M–disturbed), or disturbed. Their headline result is a relaxation fraction of $43\%^{+9}_{-9}$ for the strong-lensing sample. Here we test, cluster by cluster, whether our kinematic dynamical-state classification agrees with this independent X-ray classification.

### 7.1. *Cross-matched sub-sample*

We cross-matched the Gassis et al. (2026) table to our cluster results by parsed J2000 name core (JHHMM±DDMM), verifying each match by position. 16 clusters appear in both the Gassis et al. (2026) sample and our $N_{\rm mem} \geq 30$ DS-eligible sample; lowering the membership threshold to $N \geq 20$ adds two more (SDSS J1456+5702 and SDSS J1110+6459), for 18 overlap clusters. This overlap set is our Sample 1 and is the basis for the comparison; Table 9 lists each cluster with its kinematic properties and the Gassis et al. (2026) classification. (RCS2 J0327−1326, which has spectroscopy in our catalog, is *not* in the X-ray sample and therefore cannot enter this comparison.) For the binary comparison we collapse the four X-ray classes to *relaxed* (relaxed + M–relaxed) and *disturbed* (M–disturbed + disturbed), matching our binary kinematic classification (*disturbed* if $p_{\rm DS} \leq 0.05$, else *relaxed*).

### 7.2. *Agreement between the kinematic and X-ray classifications*

The central result is that the two classifications agree only 5/16 = 31% of the time at $N \geq 30$ (Figure 7). This level of agreement is essentially unchanged across membership cuts: 5/18 (28%) at $N \geq 20$, 5/17 (29%) at $N \geq 25$, 5/16 (31%) at $N \geq 30$, and 4/15 (27%) at $N \geq 35$. It is therefore not an artifact of a particular threshold or of small-number statistics at the boundary. The 2 × 2 contingency of the 16 clusters is

| | X-ray disturbed | X-ray relaxed |
|---|---|---|
| Kinematic disturbed | 2 | 5 |
| Kinematic relaxed | 6 | 3 |

so 5 clusters lie on the agreeing diagonal and 11 off it, split between both off-diagonal cells. Consistent with this weak correspondence, the kinematic substructure significance and the X-ray morphology parameter show no correlation across the overlap sample: the Spearman rank correlation between $-\log_{10} p_{\rm DS}$ and $M$ is $\rho = 0.18$ ($p = 0.51$).

The disagreement is asymmetric, but the asymmetry depends on the membership aperture. With velocity-only membership, of the 7 overlap clusters in which the DS test detects substructure only 2 are also classified disturbed in X-rays (SDSS J2243−0935 and SDSS J0851+3331), while the other 5 (SDSS J1038+4849, SDSS J1050+0017, SDSS J1152+0930, SDSS J1439+1208, and SDSS J1336−0331) appear X-ray-relaxed despite several having very high kinematic significance ($p_{\rm DS} < 0.001$). On its own this would suggest that the DS test picks up mergers oriented along

**Table 9.** Phase-space versus X-ray dynamical state for the Sample 1 overlap clusters (in both our $N_{\rm mem} \geq 30$ DS-eligible sample and the Chandra Strong Lens Sample of Gassis et al. (2026)), sorted by $p_{\rm DS}$.

| Cluster | $N_{\rm mem}$ | $\sigma_v$ (km s$^{-1}$) | $p_{\rm DS}$ | Phase-space class | X-ray class (Gassis+26) | $M$ | Agree? |
|---|---|---|---|---|---|---|---|
| SDSS J2243−0935 | 134 | 1358 | $<0.001$ | disturbed | disturbed | −1.9 | Y |
| SDSS J1439+1208 | 69 | 1833 | $<0.001$ | disturbed | M–relaxed | +0.1 | N |
| SDSS J1152+0930 | 88 | 1452 | $<0.001$ | disturbed | M–relaxed | +0.3 | N |
| SDSS J1050+0017 | 107 | 1860 | $<0.001$ | disturbed | relaxed | +1.1 | N |
| SDSS J1038+4849 | 128 | 1363 | $<0.001$ | disturbed | M–relaxed | +1.3 | N |
| SDSS J0851+3331 | 67 | 670 | 0.015 | disturbed | disturbed | −1.3 | Y |
| SDSS J1336−0331 | 66 | 1934 | 0.015 | disturbed | M–relaxed | +0.3 | N |
| Abell 1703 | 152 | 1367 | 0.059 | relaxed | relaxed | +1.9 | Y |
| SDSS J1329+2243 | 57 | 655 | 0.092 | relaxed | disturbed | −0.4 | N |
| SDSS J1138+2754 | 36 | 1377 | 0.221 | relaxed | disturbed | −1.2 | N |
| SDSS J1420+3955 | 37 | 1195 | 0.266 | relaxed | M–disturbed | −0.2 | N |
| SDSS J1621+0607 | 93 | 1078 | 0.363 | relaxed | disturbed | −0.6 | N |
| SDSS J1002+2031 | 50 | 1146 | 0.485 | relaxed | disturbed | −1.1 | N |
| SDSS J0928+2031 | 32 | 818 | 0.583 | relaxed | M–relaxed | +0.2 | Y |
| SDSS J0957+0509 | 39 | 1344 | 0.620 | relaxed | M–disturbed | −0.4 | N |
| SDSS J1209+2640 | 56 | 1156 | 0.858 | relaxed | M–relaxed | +0.7 | Y |

NOTE—Phase-space class: disturbed if $p_{\rm DS} \leq 0.05$, else relaxed. X-ray class from Gassis et al. (2026); for the binary comparison, relaxed+M–relaxed are grouped as relaxed and M–disturbed+disturbed as disturbed. $M$ is the combined X-ray morphology parameter (higher=more relaxed). Overall agreement: 5/16 (= 31%).

the line of sight, which leave the projected ICM smooth. The aperture test of subsection 6.5 shows that this reading does not hold for most of these clusters: in four of the five, the kinematic signal comes from galaxies beyond $R_{200}$ or from a superposed group, and within the cluster aperture only SDSS J1152+0930 remains kinematically disturbed and X-ray-relaxed.

With the aperture applied, the overall agreement is similar (5/16 within $R_{200}$ and 7/17 within 1.5 Mpc for $N \geq 20$; Table 8), so the level of agreement is not set by the aperture, but nearly all of the remaining disagreement lies in the opposite cell: 10 (9) clusters are X-ray-disturbed but show no kinematic substructure within $R_{200}$ (1.5 Mpc). The mock mergers of subsection 6.5 show that the DS test would have detected a subcluster containing a quarter of the core members and offset by $\gtrsim 1\sigma_v$ along the line of sight in several of these clusters (3.3–5.6 detections expected, none observed). The X-ray disturbances in this sample are therefore mostly not accompanied by large line-of-sight velocity substructure, which favors merger axes close to the plane of the sky, or subclusters that are too small or too advanced to stand out kinematically, in which the ICM is disrupted while the line-of-sight velocity field remains close to Gaussian.

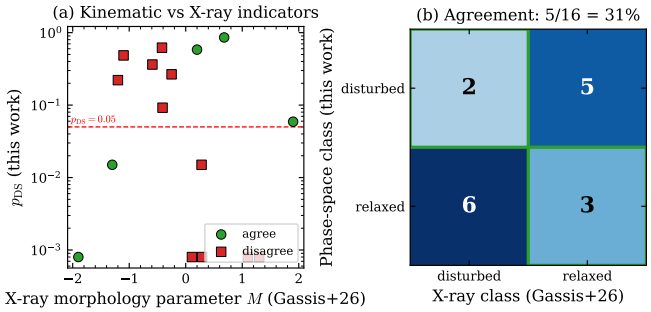



**Figure 7.** Phase-space versus X-ray dynamical state for the Sample 1 overlap clusters ($N_{\rm mem} \geq 30$, velocity-only membership; see Table 8 for the aperture-restricted comparison). *Left:* kinematic significance $p_{\rm DS}$ (this work) against the X-ray morphology parameter $M$ of Gassis et al. (2026) (higher $M$ = more relaxed); green circles agree with the X-ray binary class, red squares disagree, and the dashed line is the $p_{\rm DS} = 0.05$ threshold. *Right:* the $2 \times 2$ contingency of the binary classifications; the agreeing diagonal (outlined) holds only 5/16 clusters.

### 7.3. *Relaxation-fraction comparison*

Although the two probes disagree on which individual clusters are disturbed, they give broadly consistent *population* relaxation fractions. In the overlap sample the X-ray-relaxed fraction is 8/16 = 50% and our kine-

matically relaxed fraction is 9/16 = 56%, both consistent within Poisson uncertainties (±12%) with the $43\%^{+9}_{-9}$ relaxation fraction that Gassis et al. (2026) report for their full X-ray sample. The headline relaxed-fraction statistic is therefore reproducible across wavelengths even though the per-cluster classifications are nearly uncorrelated. This is a reminder that an agreement in aggregate fractions does not imply agreement on individual systems, and that a relaxed fraction quoted from a single probe will mis-classify a substantial minority of clusters. This agreement in aggregate fractions holds only for velocity-only membership, however. Within the cluster aperture the kinematically relaxed fraction rises to 15/16 (94%, within $R_{200}$) and 16/17 (94%, within 1.5 Mpc), against X-ray-relaxed fractions of 6/16 (38%) and 8/17 (47%) for the same clusters. The consistent population fractions above are thus partly a coincidence of the kinematic signal coming from the cluster outskirts.

### 7.4. *Worked example: SDSS J2243−0935*

SDSS J2243−0935 is one of the two clusters on which the two probes agree, and it illustrates the cleanest case of a jointly detected merger. Gassis et al. (2026) note that the optical pair associated with SDSS J2243−0935 is in a state where the "formerly separate ICMs [are] fully interacting," and they exclude it from several of their morphological analyses on the grounds that the single remaining ICM no longer permits a clean parameter measurement. This X-ray classification of a major merger in progress is consistent with our kinematic analysis (Section 6.3), which detects substructure at $p_{\rm DS} < 0.001$ and resolves a BIC-preferred two-component kinematic structure whose high-$\delta_i$ cores span $\sim$5000 km s$^{-1}$ in line-of-sight velocity. The independent agreement of optical galaxy kinematics and X-ray ICM morphology for this system validates the pipeline on a known merger. As noted in subsection 6.3, however, the kinematic detection depends on members beyond $R_{200}$ ($p_{\rm DS} = 0.16$ within $R_{200}$), so the two probes agree for this cluster with velocity-only membership but not within the cluster aperture.

## 8. DISCUSSION

### 8.1. *Selection effects in strong-lensing-selected samples*

A motivating question for this work is whether selecting clusters by the presence of giant arcs biases the population toward unusual dynamical states. Our kinematic substructure fraction depends strongly on the membership aperture. With velocity-only membership it is $f_{\rm sub} = 13/35 = 0.37 \pm 0.08$ at $p_{\rm DS} \leq 0.05$ (falling to $0.31 \pm 0.09$ if the bootstrap-seeded clusters are excluded; subsection 8.2), comparable to DS-substructure fractions reported for mass-selected and optically-selected cluster samples, which typically span ∼30–50% depending on the membership depth and the $p_{\rm DS}$ threshold adopted (e.g. Dressler & Shectman 1988; Pinkney et al. 1996). Within $R_{200}$ or 1.5 Mpc, however, it is only 0.11–0.15 (subsection 6.5), so comparisons with published fractions need to match the spatial extent of the membership as well as $N$ and the $p_{\rm DS}$ threshold. The relaxed fraction we infer kinematically in the X-ray overlap sample with velocity-only membership (56%, subsection 7.3) is consistent with both the 50% X-ray-relaxed fraction in the same overlap clusters and the $43\%^{+9}_{-9}$ that Gassis et al. (2026) measure for their full Chandra strong-lensing sample. Taken together, these fractions provide no evidence that strong-lensing selection drives the population toward an anomalously high merging fraction; the strong-lensing sample looks dynamically ordinary at the level our statistics can probe. The more informative result is the *per-cluster* comparison: the two probes agree on individual classifications for only 5 of 16 overlap clusters (31%), both with velocity-only membership and within $R_{200}$, and once membership is restricted to the cluster the disagreement is dominated by X-ray-disturbed clusters that lack line-of-sight kinematic substructure (subsection 7.2).

The velocity-only DS signal is nonetheless informative about the environment of strong-lensing clusters. In several fields it traces groups superposed along the line of sight a few thousand km s$^{-1}$ from the cluster (SDSS J1050+0017, SDSS J1336−0331) and structure at 2–6 Mpc. Correlated structure along the line of sight can add to the lensing efficiency of a cluster, and its incidence toward SGAS lenses was studied by Bayliss et al. (2014); together with the orientation bias expected for strong-lensing selection (Meneghetti et al. 2014), this makes the separation of core substructure from line-of-sight structure, as done here with the aperture comparison, essential when interpreting kinematic substructure in lensing-selected samples.

### 8.2. *Limitations*

Our analysis has several limitations that should be borne in mind when interpreting the results.

*Spectroscopic completeness* varies across the sample, from $N_{\rm mem} = 30$ (the DS test threshold) to several hundred in the deepest fields. At low $N_{\rm mem}$ the DS test loses statistical power (Pinkney et al. 1996) and the GMM is prone to over-fitting; to mitigate the latter we require the decisive $\Delta$BIC $\geq 6$ threshold in the GMM model selection (subsection 5.5).

*Heterogeneous catalog construction.* Although we have carefully cross-matched and deduplicated the input catalogs, the underlying selection functions of NED, Sharon2020, Bayliss2011, Boschin2020, and Carrasco2017 are all different. Our source-catalog robustness check (Section 6.4) explicitly tests for catalog-driven artifacts in the substructure signal.

*Bootstrap-seeded cluster redshifts.* For 9 of the 54 clusters meeting the $N \geq 15$ threshold no published spectroscopic biweight redshift exists, and we bootstrap the pre-filter seed $z_{\rm cl}$ from a KDE peak on the raw field (subsection 5.1). These nine clusters are SDSS J1050+0017, SDSS J1439+1208, SDSS J0146−0929, SDSS J1336−0331, SDSS J1329+2243, SDSS J1055+5547, SDSS J0108+0624, SDSS J1110+6459, and SDSS J1723+3411; six of them reach the $N \geq 30$ DS threshold. The reported $z_{\rm bi}$ and $\sigma_v$ are computed independently of the seed by iterative biweight convergence, but the systematic uncertainty associated with the pre-filter centring is larger than for clusters whose seed comes from a vetted publication.

Within the bootstrap subset, 4/6 DS-eligible clusters show substructure ($f_{\rm sub}^{\rm bootstrap} = 0.67 \pm 0.19$), compared with 9/29 in the published-$z_{\rm cl}$ subset ($f_{\rm sub}^{\rm published} = 0.31 \pm 0.09$). The four bootstrap detections all have $\sigma_v > 1{,}700\,{\rm km\,s^{-1}}$, suggesting either (i) that clusters without a published cluster redshift are preferentially those for which prior observers did not derive clean dynamics because the systems are morphologically complex mergers, or (ii) that the bootstrap pre-filter retains line-of-sight interlopers that inflate $\sigma_v$ and $\delta_i$. The two interpretations are not mutually exclusive. Excluding the bootstrap subset gives $f_{\rm sub} = 9/29 = 0.31 \pm 0.09$ for the published-$z_{\rm cl}$ DS-eligible sample, slightly below the inclusive value reported in subsection 6.2. The aperture test of subsection 6.5 favors the second interpretation: none of the four bootstrap-seeded detections (SDSS J1050+0017, SDSS J1439+1208, SDSS J0146−0929, and SDSS J1336−0331) is significant within 1.5 Mpc, and the two that retain enough members within $R_{200}$ are not significant there either.

*Membership aperture.* Our fiducial tables use velocity-only membership, which includes galaxies well beyond $R_{200}$ in the most widely surveyed fields; subsection 6.5 quantifies the effect of restricting it. Estimating $R_{200}$ from $\sigma_v$ is circular when $\sigma_v$ is itself inflated by contaminants (SDSS J1336−0331 is bistable for this reason), which is why we also report a fixed 1.5 Mpc aperture; the two give consistent results. An aperture based on independent lensing or X-ray masses would remove this circularity.

*Projection effects.* The DS test and the GMM both work in projected (RA, Dec, $v_{\rm LOS}$) space, and consequently are insensitive to plane-of-sky merger geometries, in which the merging halos have little relative line-of-sight velocity. The mock mergers of subsection 6.5 quantify this sensitivity for each overlap cluster. The X-ray morphological tests of Gassis et al. (2026) are complementary in this respect and detect plane-of-sky mergers more easily; this is the underlying physical motivation for the joint comparison in Section 7.

## 9. SUMMARY AND CONCLUSIONS

We have presented a new spectroscopic redshift catalog and a dynamical substructure analysis for a sample of strong-lensing-selected galaxy clusters from the SDSS Giant Arcs Survey. Our main results are as follows:

1. We have obtained new IMACS/Magellan-I (Baade) multi-object spectroscopy in the fields of 6 SGAS clusters and re-measured every spectrum from scratch with MARZ, yielding 980 redshifts with QOP $\geq 2$, 912 of them with QOP $= 4$; a second independent MARZ pass reproduces >95% of them (Section 3).

2. Merged with literature catalogs and NED, we provide a homogenized master spectroscopic catalog of 9,176 pre-deduplication entries (7,543 unique objects after 1.″0 cross-matching) across 74 cluster fields, of which 54 have $N \geq 15$ members for biweight dynamics and 35 have $N \geq 30$ members for the DS substructure test (Section 4).

3. We have implemented an end-to-end pipeline that performs iterative biweight membership selection, biweight dynamics with jackknife uncertainties, the DS substructure test with Monte-Carlo calibration, and GMM-based substructure verification, with reproducible code released alongside this paper (Section 5).

4. We identify kinematic substructure ($p_{\rm DS} \leq 0.05$) in 13 of the 35 DS-eligible clusters, yielding a strong-lensing-sample kinematic substructure fraction of $f_{\rm sub} = 0.37 \pm 0.08$, dropping to $6/35 = 0.17 \pm 0.06$ under the more conservative $p_{\rm DS} < 0.01$ threshold (Section 6.2). Restricting membership to an iterated $R_{200}$ or to 1.5 Mpc lowers this to 0.11–0.15; mock mergers on the observed galaxy positions show that the drop is not a loss of statistical power, so most velocity-only detections trace structure beyond $R_{200}$ or superposed groups (Section 6.5).

5. We present SDSS J2243−0935 as a worked example of a high-significance near-line-of-sight major merger, detected with $p_{\rm DS} < 0.001$ in the DS test and confirmed by a BIC-preferred two-component GMM; the bulk components are separated by $\sim 1{,}380\,{\rm km\,s^{-1}}$ and the high-$\delta_i$ cores by $\sim 5000\,{\rm km\,s^{-1}}$ in line-of-sight velocity (Section 6.3). Its DS detection depends on members beyond $R_{200}$ ($p_{\rm DS} = 0.16$ within $R_{200}$).

6. Cross-matching the DS-eligible sample against the X-ray morphological classifications of Gassis et al. (2026), we find that the kinematic and X-ray dynamical-state classifications agree for only 5 of the 16 overlap clusters (31% at $N \geq 30$, stable across membership cuts), and that of the 7 kinematically substructured clusters only 2 are also X-ray-disturbed (Section 7). The aggregate relaxed fractions nonetheless agree (56% kinematic vs. 50% X-ray in the overlap, both consistent with the 43% of Gassis et al. 2026). With membership restricted to the cluster aperture the agreement is unchanged (5 of 16 within $R_{200}$), but the disagreement reverses: only SDSS J1152+0930 is kinematically disturbed and X-ray-relaxed, while none of the 10 X-ray-disturbed clusters shows kinematic substructure within $R_{200}$, which our mocks show is unlikely if their mergers carried large line-of-sight velocity offsets (Sections 6.5–7.4).

7. SDSS J1152+0930 shows kinematic substructure that is robust to every test we applied, at every aperture, while appearing moderately relaxed in X-rays, making it the strongest candidate for a merger close to the line of sight in the sample (Section 6.5).

## 10. FUTURE WORK

Several extensions of this analysis are natural next steps. First, deeper and more uniform multi-object spectroscopy of the remaining SGAS fields would raise more clusters above the $N \geq 30$ DS threshold and tighten the per-cluster comparison. Second, the aperture comparison of subsection 6.5 should be repeated with $R_{200}$ set from independent lensing or X-ray masses, and the line-of-sight structure that dominates the velocity-only DS signal should be characterized directly. Third, a full per-cluster, per-source-label re-run of the DS test (subsection 6.4) and a calibrated per-object redshift error budget from repeat observations should accompany the machine-readable catalog release. Finally, comparing the substructure and relaxed fractions measured here to mass-selected samples such as CHEX-MATE (CHEX-MATE Collaboration et al. 2021) would place the strong-lensing selection on a quantitative footing relative to X-ray- and SZ-selected populations.

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*Facilities:* Magellan:Baade (IMACS), HST (WFC3), CXO (ACIS-I, ACIS-S)

*Software:* Astropy (Astropy Collaboration et al. 2022), astroquery (Ginsburg et al. 2019), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), scikit-learn (Pedregosa et al. 2011), Matplotlib (Hunter 2007), Marz (Hinton et al. 2016), RVSAO (Kurtz & Mink 1998), COSMOS (Kelson 2003)

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